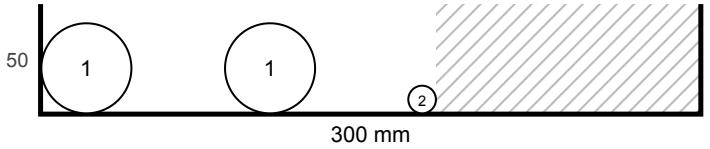
Villa Complex 415V LV Network — Cable tray schedule

REV —

Perforated tray (HDG): default spare 25 %, single layer, spacing 1 D, depth 50 mm, max width 600 mm.

Route A — → — · 20 m

No.	From (panel)	To	Cable	Path	Qty	OD (mm)	kg/m
1			4C × 95 mm²	A only	2	41.7	5.4
2			1C × 50 mm²	A only	1	13.2	0.5



Cross-section to scale — hatched = spare

Single layer, spacing 1 D: Σ D 96.6 + spacing 83.4 = 180 mm; + 25 % spare → required 225 mm.

Selected tray: 300 × 50 mm — spare 67 %, cable weight 11.3 kg/m, grouping factor 0.98 (3 per tier, spaced), bend radius ≥ 340 mm

Summary

Route	From	To	Panels	Cables	Required (mm)	Tray W × D (mm)	Spare	Grouping	kg/m	Length (m)	Status
A				3	225	**300 × 50**	67 %	0.98	11.3	20	PASS

Tray BOQ

Size W × D (mm)	Tray (m)	Bends	Tees	Reducers	Risers	Supports	Coupler sets	Bend radius ≥ (mm)	Routes
300 × 50	20	2	0	0	0	15	6	340	A

Tiers counted in the BOQ. Grouping factors: IEC 60364-5-52 Table B.52.20 (trays) / B.52.17 (bunched), typical values. Cable diameters: DUCAB 0.6/1 kV Cu XLPE/SWA/PVC — diameters and bending radii from DUCAB; weights typical.